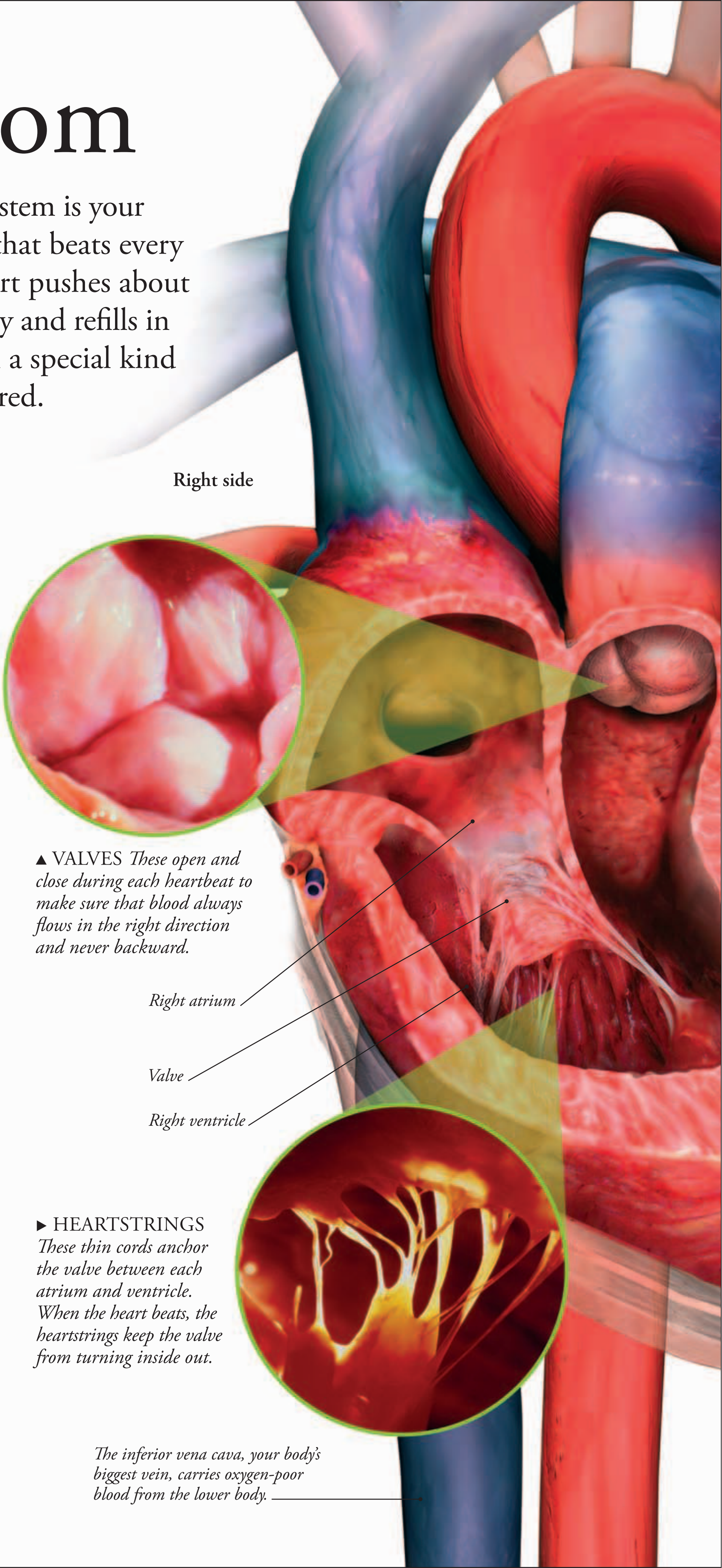
Engine room

At the center of your circulatory system is your heart, a fist-sized powerful pump that beats every second. With every beat, your heart pushes about a cupful of blood around your body and refills in time for the next beat. Made from a special kind of muscle, your heart never gets tired.

INSIDE THE HEART

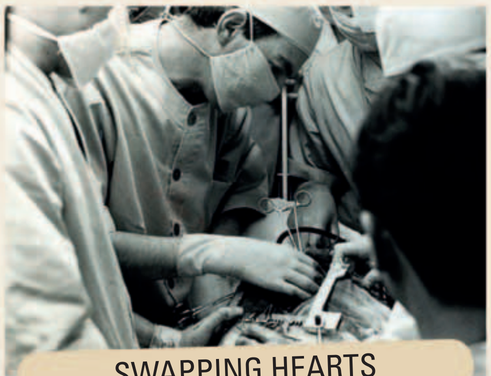
Your heart lies in the middle of your chest, sandwiched between your two lungs and protected by the ribcage. It has two sides, each with a smaller, upper space (the atrium), and a lower, larger one (the ventricle). The right side of your heart pumps blood to your lungs, while the left side receives blood from your lungs and pumps it to every part of your body.



▲ VALVES *These open and close during each heartbeat to make sure that blood always flows in the right direction and never backward.*

▶ HEARTSTRINGS *These thin cords anchor the valve between each atrium and ventricle. When the heart beats, the heartstrings keep the valve from turning inside out.*

The inferior vena cava, your body's biggest vein, carries oxygen-poor blood from the lower body.



SWAPPING HEARTS

Before the 1960s, heart transplants were thought to be impossible because of the high risk of death. But in 1967, history was made when a pioneering South African surgeon, Christiaan Barnard (center), carried out the world's first successful heart transplant. He removed the damaged heart from his patient and replaced it with a healthy heart from a donor. Today, heart transplants are performed all the time.